

Experimental Report 04

Adaptive Quantile Calibration and Error-Safe RUL Prediction

Remaining Useful Life prediction from multivariate sensor time series

Approach	Best MAE	Best R2	Best method
quantile RF	8.92 cycles	0.91	ensemble, cap 140

Quantile experiment setup

- The previous baseline used the mean prediction of 400 Random Forest trees and then subtracted one fixed shift for every engine.
- The goal was to reduce MAE and dangerous RUL overestimation, especially in the near-failure region, without moving to heavy neural architectures.
- The main change was to replace the fixed mean-plus-shift rule with adaptive lower-quantile calibration over the distribution of tree predictions.
- Candidate settings were selected not only by MAE, but also by overestimation share and near-failure / warning-zone behavior.

Main tested configurations

Method	MAE	RMSE	R2	Over-est.	Near-fail MAE	Note
Mean + shift baseline	9.45	13.16	0.90	48%	2.58	baseline line
Quantile (q=0.35, shift=0.5)	9.27	13.81	0.89	37%	2.43	first adaptive breakthrough
Best quantile by MAE (q=0.40)	9.10	13.30	0.90	48%	2.41	best single-quantile MAE
Best quantile by fitness (q=0.35)	9.27	13.81	0.89	37%	2.43	best safety balance
Best quantile ensemble (cap=140)	8.92	12.76	0.91	47%	2.38	best MAE in this stage
Best cap-refined (q=0.40, cap=140)	9.05	12.98	0.90	50%	2.43	cap refinement check
Best piecewise quantile	9.01	13.00	0.90	49%	2.39	zone-based calibration
Best residual correction	9.58	13.60	0.89	43%	2.46	worse than quantile

Best tested configuration: quantile ensemble with cap=140. The model reached MAE 8.92, RMSE 12.76 and R2 0.91.

Result interpretation

- The best variant reduced MAE from 9.45 to 8.92 cycles and improved RMSE from 13.16 to 12.76.
- Lower quantiles gave more conservative predictions, sharply reducing overestimation at q=0.35, while the lowest MAE came from the quantile ensemble with cap refinement.

Quantile Calibration and Refinement

Quantile prediction logic

Instead of averaging the predictions of all Random Forest trees, the model uses a lower quantile of the per-tree prediction distribution. This makes the final RUL estimate more cautious when the ensemble itself indicates uncertainty.

- Quantile grid: q from 0.10 to 0.50.
- Optional shift values: 0 and 0.5 cycles.
- Selection metric: composite fitness combining MAE, overestimation penalty, and near-failure / warning-zone behavior.

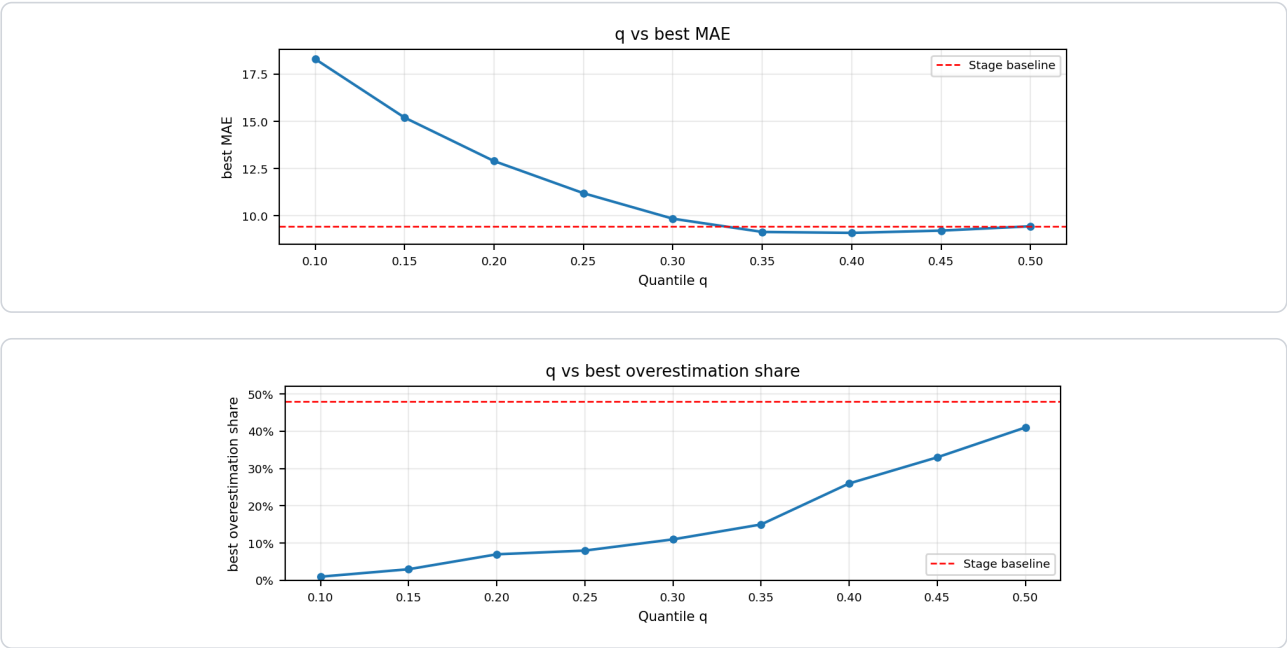


Figure 1. Quantile sweep. Lower quantiles strongly reduce dangerous overestimation, while the MAE minimum is reached near $q=0.35-0.40$.

Refinement variants

Variant	What it changes	Role in this stage
Cap refinement	Searches RUL cap values such as 120, 125, 130, 140, 150 together with q and shift.	Improved high-RUL control; cap=140 became part of the best ensemble.
Piecewise quantile	Uses different quantiles for near-failure, warning and long-RUL zones.	Reached MAE 9.01 and near-failure MAE 2.39.
Uncertainty-based rule	Moves to a lower quantile when the tree prediction spread is high.	Useful as a safety-oriented calibration idea.
Quantile ensemble	Blends quantile predictions, for example combining $q=0.35$ with the mean prediction.	Best result: MAE 8.92, RMSE 12.76, R2 0.91.

Residual correction check

A second Random Forest was trained to predict residual error using internal validation splits. It was operationally valid, but official-test MAE was worse than the best quantile variants, so it was kept as a checked alternative rather than the main approach.

Prediction Behavior and Error Direction

True versus predicted RUL

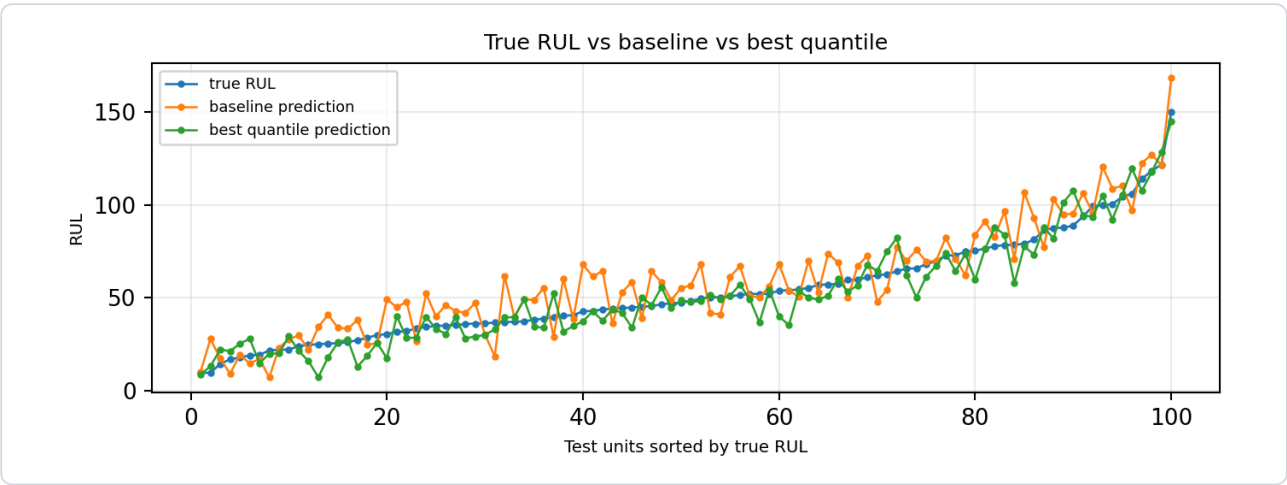


Figure 2. The best quantile prediction follows the main target structure while reducing the most dangerous upward bias relative to the mean-plus-shift baseline.

Error sign by test unit

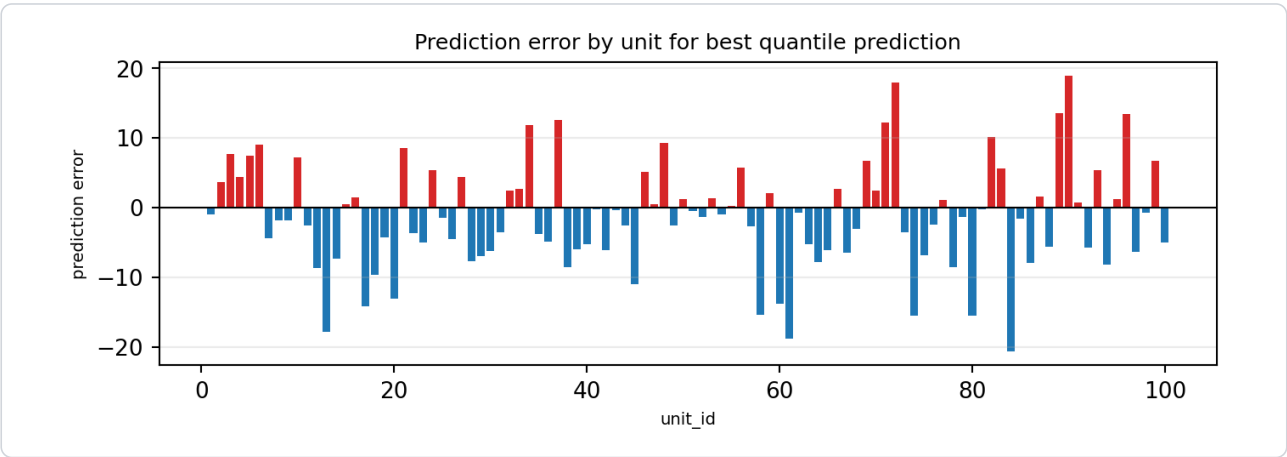


Figure 3. Positive errors correspond to RUL overestimation. The calibration step is evaluated not only by MAE, but also by how often it moves predictions into the safer underestimation direction.

Behavior summary

Aspect	Observed behavior	Interpretation
High-RUL units	The largest deviations remain more visible at the long-life end of the test set.	Cap refinement directly limits unrealistically high predictions.
Near-failure units	Near-failure MAE improved from 2.58 to 2.38 in the best ensemble.	The safety direction matters more than a small aggregate gain.
Error direction	q=0.35 reduced overestimation share from 48% to 37%.	Lower quantiles work as an error-safe calibration mechanism.

Zone Diagnosis and Next Direction

Baseline zone diagnosis

Zone	Engines	MAE	Overestimation share	Interpretation
Near failure (≤ 30)	21	2.58	57%	low absolute error, but safety direction still matters
Warning (31-70)	17	7.98	71%	highest overestimation tendency, important calibration zone
All test	100	9.45	48%	baseline reference level

Connection to later architecture research

- The literature review suggests a path from CNN-LSTM filtering of noisy sensor history to Attention-LSTM selective focus and Transformer-based global cross-sensor relations.
- Reported reference points in the presentation: CNN-LSTM around MAE 9.70, Attention-LSTM around MAE 8.95, and Transformer-style approaches around MAE 8.20.
- The main methodological gap is that many strong architectures optimize aggregate MAE but do not explicitly distinguish dangerous RUL overestimation from safer underestimation.

Architecture comparison

Direction	Core idea	Risk for this project
DCNN-LSTM	CNN filters noisy local sensor patterns; LSTM models degradation trend.	Training complexity and two-network pipeline.
Attention-LSTM	Adds cycle importance weights and focuses on degradation-relevant intervals.	Needs careful validation to avoid unstable attention on small data.
Transformer	Models global correlations across sensors and the whole observation window.	High overfitting risk without strict regularization and enough data.

Report summary

Main update	fixed shift replaced by adaptive quantile calibration over Random Forest tree predictions
Best result	quantile ensemble with cap=140: MAE 8.92, RMSE 12.76, R2 0.91
Safety behavior	q=0.35 reduced overestimation share from 48% to 37%, while the ensemble gave the lowest MAE
Residual branch	tested as an alternative, but weaker than quantile calibration on official-test MAE
Next direction	move from manual quantile calibration toward selective attention / transformer-style models with explicit direction-aware error control